

Application: 4877434909

Rose Bruffaerts - rose.bruffaerts@med.kuleuven.be
MISTI Global Seed Funds



Fund Selection

Completed - Sep 4 2019

Fund Selection

Project Title

Please provide a descriptive name your project

Towards a neural marker of complex meaning construction

MIT Administrative Officer (AO)

If you are the international applicant, and you do not have this information, you should ask your MIT collaborator. A list of AOs can be found here for reference. Do not put in the name of the MISTI Program Manager.

<i>First name</i>	<i>Gayle</i>
<i>Last name</i>	<i>Lutchen</i>
<i>E-mail</i>	<i>gwolf@MIT.EDU</i>

MIT Fiscal Officer (FO) or Departmental Assistant/Support Staff

Please list the individual within your office/department that processes your travel reports and other financial processes. A list of FOs can be found [here](#) for reference.

If you are the international applicant, and you do not have this information, you should ask your MIT collaborator. Do not put in the name of the MISTI Program Manager.

First name	Donna
Last name	Wells
E-mail	donnab@mit.edu

MIT Fund Center Number

This information may be obtained from the MIT collaborator's Administrative Officer or Fiscal Officer.

207698

MIT Profit Center Number

This information may be obtained from the MIT collaborator's Administrative Officer or Fiscal Officer.

400600

Please select which fund you are applying to:

MIT - KU Leuven Seed Fund

University-specific funds

For university-specific funds, faculty members from that respective institution should direct their applications to those funds.

These include:

- *Ben-Gurion University of the Negev Seed Fund*
- *MIT - KU Leuven Seed Fund*
- *MIT-Belgium - Universite catholique de Louvain Seed Fund*
- *MIT-Chile - Universidad de Santiago de Chile Seed fund*
- *MIT-Chile - Universidad Adolfo Ibanez Fund*
- *MIT-Colombia Universidad de los Andes*
- *MIT-Germany - FAU Seed Fund*
- *MIT-Germany - University of Regensburg Seed Fund*
- *MIT-Germany - University of Stuttgart Seed Fund*
- *MIT-Korea KAIST Seed Fund*
- *MIT-Peru UTEC Seed Fund*

If you are unsure which fund you should apply to, please visit the Available Funds page for further information.

To submit your response and move on to the application, click on the green "Mark as Complete" button below.



Fill out MISTI Global Seed Funds Application

Completed - Sep 12 2019

MISTI Global Seed Funds Application

Page 1: MIT Applicant and Fund Information

Project title:

4877434909

To edit your project title, go back to the "Fund Selection" task in the task bar on the left-hand side of this screen.

Proposal field - your proposal will be reviewed by someone in this field (choose up to 2)

Field 1	Brain and Cognitive Sciences
Field 2	Linguistics and Philosophy

Please list specific subfield (energy, environmental science, particle physics, robotics, etc.)

Neurolinguistics

Number of MIT Applicants (Faculty)

2

MIT faculty or research scientist applicant:

GSF is open to all MIT faculty members and research staff with principal investigator privileges (principal research scientists and senior research scientists). If you have questions about your PI status, please consult your department.

First name	Evelina
Last name	Fedorenko
Status	Assistant Professor
Title	Professor
Department	Brain and Cognitive Sciences (Course 9)
School	School of Science
Lab or Center	EvLab
MIT E-mail	evelina9@mit.edu
Does this applicant have PI (Principal Investigator) Status?	Yes
Are they the main PI on this project?	Yes

GSF is open to all MIT faculty members and research staff with principal investigator privileges (principal research scientists and senior research scientists). If you have questions about your PI status, please consult your department.

MIT faculty or research scientist applicant 2:

<i>First name</i>	<i>Dimitrios</i>
<i>Last name</i>	<i>Pantazis</i>
<i>Status</i>	<i>Principal Research Scientist</i>
<i>Title</i>	<i>Research Scientist</i>
<i>Department</i>	<i>Brain and Cognitive Sciences (Course 9)</i>
<i>School</i>	<i>School of Science</i>
<i>Lab or Center</i>	<i>MEG lab</i>
<i>MIT E-mail</i>	<i>pantazis@MIT.EDU</i>
<i>Does this applicant have PI (Principal Investigator) Status?</i>	<i>Yes</i>
<i>Are they the main PI on this project?</i>	<i>No</i>

GSF is open to all MIT faculty members and research staff with principal investigator privileges (principal research scientists and senior research scientists). If you have questions about your PI status, please consult your department.

Number of International collaborators

2

International collaborator applicant 1

<i>First name</i>	<i>Rose</i>
<i>Last name</i>	<i>Bruffaerts</i>
<i>Title</i>	<i>Postdoctoral research fellow</i>
<i>Institution</i>	<i>KU Leuven</i>
<i>Department</i>	<i>Neurosciences</i>
<i>Lab or Center</i>	<i>Laboratory for Cognitive Neurology</i>
<i>Street address</i>	<i>Herestraat 49</i>
<i>City and postal code</i>	<i>Leuven 3000</i>
<i>Country</i>	<i>Belgium</i>
<i>E-mail</i>	<i>rose.bruffaerts@kuleuven.be</i>
<i>Research Group Website</i>	<i>http://med.kuleuven.be/lcn</i>

International collaborator applicant 2

<i>First name</i>	<i>Rik</i>
<i>Last name</i>	<i>Vandenberghe</i>
<i>Title</i>	<i>Professor</i>
<i>Institution</i>	<i>KU Leuven</i>
<i>Department</i>	<i>Neurosciences</i>
<i>Lab or Center</i>	<i>Laboratory for Cognitive Neurology</i>
<i>Street address</i>	<i>Herestraat 49</i>
<i>City and postal code</i>	<i>Leuven 3000</i>
<i>Country</i>	<i>Belgium</i>
<i>E-mail</i>	<i>rik.vandenberghe@uzleuven.be</i>
<i>Research Group Website</i>	<i>http://med.kuleuven.be/lcn</i>

Please take note, your proposal will be disqualified if you have any outstanding reports.

Please sub your report via this form. If you have any questions, please contact misti-gsf@mit.edu.

MIT faculty or research scientist applicant: Does at least one of your MIT applicants have PI (Principal Investigator) Status?

GSF is open to all MIT faculty members and research staff with principal investigator privileges (principal research scientists and senior research scientists). If you have questions about your PI status, please consult your department.

Yes

What non US country is involved in this project?

Belgium

Does your project involve more than one non US country?

No

Brief abstract

Provide a brief summary of the proposed project written for an audience of non-experts. This summary may be used in public promotional materials. (limited to 250 words)

Human language is capable of conveying a vast number of ideas. At the core of this expressive power is the principle of semantic compositionality: the meaning of an utterance is determined by the meanings of its parts. How do we infer complex ideas from others' utterances? In a recent study, using intracranial recordings from the surface of the human brain (electrocorticography or ECoG), we found that neural activity increases monotonically over the course of a sentence across the language system. Having ruled out a number of alternative explanations of this effect in terms of generic attention and cognitive load, we argued that this response increase reflects the increasing complexity of the evolving representation of the sentence meaning and is thus a candidate neural marker of complex meaning construction. Although ECoG is a powerful method with high spatial and temporal resolution, only a few patients undergo this procedure, making data collection slow, and the coverage of the brain is limited, making it difficult to understand the language system as a whole. Here, we propose to use magnetoencephalography (MEG) – a sensitive non-invasive technique which provides whole-brain coverage and can be used in neurotypical individuals – to probe sentence comprehension, including complex meaning construction, in greater detail. In particular, building on our pilot collaborative work, we here aim to validate the first language “localizer” for MEG, and identify a neural marker of sentence construction, laying an important foundation for discovering the precise sequence of computational and neural events that underlie sentence interpretation.

Description of collaboration (if applicable)

Provide a brief history of the collaboration and describe how your work is complementary. (limited to 250 words)

Our collaboration started in 2018, when we decided to combine efforts to develop a functional “localizer” for language-responsive areas for MEG. Fedorenko’s lab has pioneered the functional localization approach for language in fMRI. Unlike the traditional approach, where individual brains are averaged together in the common space, this approach takes into account the well-established inter-individual variability in the precise locations of functional areas, conferring higher sensitivity and functional resolution, and allowing for straightforward comparisons of findings across individuals, studies, and labs as needed to establish a cumulative research enterprise. Furthermore, understanding inter-individual variability in the neurotypical population can shed light on the nature of pathologies in neurological disorders. In terms of methodological expertise, Fedorenko brings her knowledge of individual-subject analyses, and Bruffaerts – her MEG expertise (training at the University of Cambridge, UK/MRC Cognition & Brain Sciences Unit as a postdoctoral fellow in 2016-17). On the theoretical side, Fedorenko has been investigating language comprehension for the last 17 years, and Bruffaerts (a neurologist) has spent her graduate and postdoctoral years studying semantic processing of single concepts, in both neurotypical individuals and individuals with neurodegenerative disorders. MEG data on the functional localizer paradigm adapted from Fedorenko’s fMRI work have already been collected in Dutch and English. Bruffaerts visited the Fedorenko Lab in April-May 2019 (funded by the Research Foundation Flanders) to set up a common pre-processing and analysis pipeline with the help of Pantazis (MIT) and perform preliminary data analyses. The second visit is tentatively planned for spring 2020, pending funding.

Description of the proposed project

Describe in detail the proposed project. Funding may be used to cover travel and meeting costs to facilitate international projects and collaboration. Funds may not be used to cover salaries or materials and long-term stays. Please include any technical information that will help scientific reviewers in your field evaluate the proposal. Please review the guidelines and eligibility from our website. (limited to 1500 words)

Sentence comprehension occurs rapidly, and can only be understood at a mechanistic level by discovering the precise sequence of underlying neural events. This project will illuminate the computations that support sentence interpretation, with a focus on complex meaning construction, by combining a) magnetoencephalography (MEG), a technique with high temporal resolution and whole-brain coverage, with b) the power of individual-subject analyses, which take into account the well-established variability in the precise locations of functional areas. MEG has been gaining popularity in cognitive neuroscience but is still vastly underused in language research. Individual-subject analyses

yield increases in sensitivity, while at the same time improving the interpretability of the results and facilitating comparisons across studies and labs. Thus this combined approach, which has only been previously used in the domain of vision (Liu et al., 2001), holds substantial promise for language research.

Fedorenko et al. (2010) developed a robust and replicable approach to identifying language-responsive brain regions in individual participants using fMRI. They then established that these language-responsive areas are highly selective for processing linguistic input (e.g., Fedorenko et al., 2011) and support both single-word comprehension and combinatorial processing (e.g., Fedorenko et al., 2012, 2019), placing constraints on the computations that take place in these circuits. However, a major limitation of fMRI is its low temporal resolution (seconds). In 2016, Fedorenko and colleagues extended the functional localization approach to electrocorticography (ECoG) and discovered an intriguing phenomenon: an increase in gamma power (70-170Hz) over the course of a sentence. Having ruled out a number of alternative explanations of this effect in terms of generic attention and cognitive load, they argued that this increase reflects the increasing complexity of the evolving representation of the sentence meaning and is thus a candidate neural marker of complex meaning construction. In principle, this discovery opens the door to probing real-time linguistic computations related to semantic interpretation using novel paradigms, but ECoG is invasive and data are scarce. As a result, we propose to use MEG, which can track neural activity at a millisecond resolution across the whole brain non-invasively in neurotypical participants, to tackle the fundamental question of how meaning is extracted from the linguistic signal.

The specific aims of the project are: 1) to validate the first individual-participant language “localizer” for MEG (in Dutch and English), and 2) identify a neural marker of sentence construction. Further, given the success of the functional localization approach in fMRI, which has now been adopted by dozens of labs in the US, Europe, and Asia, we expect that this approach will find wide resonance in the MEG field, and that the language localizer paradigm will become a key paradigm in MEG language research. Materials and scripts will be made available online.

Forty young right-handed neurotypical participants (20 Dutch speakers, 20 English speakers) will perform an adapted version of the language localizer paradigm (Fedorenko et al., 2010). The task consists of two conditions: reading sentences and reading sequences of nonwords (80 trials per condition, 12 words/nonwords each). The sentences > nonwords contrast targets areas that support lexico-semantic and combinatorial (semantic and syntactic) processing. The sentences are constructed so as to use a variety of syntactic structures and cover a wide range of topics. The nonwords are created by replacing some phonemes in real words to create phonotactically legal (and thus pronounceable) strings. At the end of each sentence or nonword string, participants are asked to decide whether a probe word or

nonword appeared in the preceding trial by pressing a button. This task ensures engagement, which is especially important in the nonwords condition. The materials are presented one word/nonword at a time in the center of the screen (385ms each), for the total duration of ~20 min. Continuous MEG data are recorded using a whole-head 306 channel (102 magnetometers, 204 planar gradiometers) Triux system (Elekta Neuromag, Helsinki, Finland) at the CUB Hôpital Erasme (Brussels, Belgium) for the Dutch experiment and at the Martinos Imaging Center (MIT) for the English experiment. So far, 19 Dutch-speaking volunteers (age 19-29), and 5 English-speaking volunteers (age 18-23) have participated in this experiment. Preprocessing consists of low-pass filtering and removal of eye blinks and heartbeats using ICA. The preprocessing pipeline was automated during RB's first research stay in the Fedorenko Lab (in April-May 2019).

Primary analyses will be performed in sensor space and will follow the general approach from Fedorenko et al., 2010. Processing will be performed using Brainstorm, an open-source MEG analysis toolbox which was co-developed by Pantazis. First, we will establish that using MEG, we are able to robustly identify language-responsive regions at the single-participant level and demonstrate the increase in sensitivity using the individual-subjects functional localization approach compared to a traditional group-level analysis (separate analyses will be performed for magnetometers and gradiometers). To do this, we will run the data through two parallel analysis streams. In both streams, the data in each individual will be divided in half (odd- vs. even-numbered trials, as in Fedorenko et al., 2010). We will directly evaluate the stability of the language responses within individuals across time and within each frequency band (i.e., between the odd- vs. even-numbered trials). We will do this in two ways: a) by identifying sensors that show a significant sentences>nonwords effect (at some fixed significance threshold) in each half of the data and computing the overlap between these relative to the sum of all sensors showing a significant sentences>nonwords effect; and b) by correlating the size of the sentences>nonwords effect across all sensors within each individual between the two halves of the data. Further, in the individual-subjects stream, the data from the odd-numbered trials will be used to define sensors of interest in each participant individually. In particular, we will take the 10% sensors that show the sentences>nonwords effect most robustly. In the group-analysis stream, the data from the odd numbered trials will be averaged across participants, and the top 10% of sensors will be defined based on this group map. Then, in the critical step, the responses to the sentences and nonwords conditions will be extracted from the even-numbered trials using the participant-specific sensors of interest (in the individual-subjects stream) or the group-level sensors of interest (in the group-analysis stream). The effect sizes will then be compared between these two approaches. Third, among the language-responsive sensors, we will search for sensors that show an activity increase over the course of the sentence, as we found in ECoG (see also Hultén et al., 2019, for a report of such a signal in MEG language data). Again, this search will be performed using the odd-numbered trials (exploring different frequency bands), and tested in the even-

numbered trials. Language-responsive sensors, including a subset that potentially show the sentence “build-up” effect, will be visualized as topographical maps. Finally, similar analyses will be conducted in source space (minimum norm imaging, dSPM).

Preliminary analyses (on the Dutch speakers; $n=19$) have yielded promising results: language responses (i.e., sentences > nonwords effects) were significantly larger when sensors were selected at the individual level compared to the group level (paired t-tests: magnetometers $p=0.0005$, gradiometers $p=0.00001$). The sentences > nonwords effect was stronger using the individual sensor selection vs. the group-level selection in 18/19 subjects for magnetometers, and in 17/19 for gradiometers. In a ~quarter of the subjects, using the group-level sensor selection, activity was actually higher for nonwords than for sentences (5/19 for magnetometers and 3/19 for gradiometers), clearly demonstrating that selecting sensors at the group level can largely miss language-responsive areas in a substantial proportion of participants. Finally, the sentences > nonwords effect sizes in the language-responsive sensors were compared between the odd and even trials. When selecting sensors at the individual level, the correlation between the two halves of the dataset was ~ 0.4 , whereas it was ~ 0.1 at the group level. These absolute values are were similar to the correlations reported for the fMRI version of the paradigm (Fedorenko et al., 2010, Fig 9). These preliminary results already confirm that language-responsive sensors can be robustly identified at the individual level. Thus, this approach holds substantial promise for future MEG work on language comprehension.

Next, we will focus on identifying a neural marker of complex sentence construction as described above. If this step is successful (which is likely given suggestions of this kind of an effect in prior MEG studies; Lam et al., 2016; Hultén et al., 2019), we are planning a series of experiments to probe the process of semantic composition further. In particular, we will vary the properties of the linguistic stimuli, including their length (to determine when the semantic build-up effect may asymptote), their syntactic well-formedness (e.g., see Mollica et al., 2019, for fMRI evidence that semantic dependencies are formed even when the syntactic structure of a sentence is degraded), the lexico-semantic features of the individual words and the relationships among them, etc. These MEG studies will be accompanied by parallel behavioral investigations of similar materials in patients with neurological diseases as a result of stroke or degeneration, to shed light on the language architecture by understanding the changes in language processing following brain damage.

Exchanges and activities planned, including student participation

Describe in detail the exchanges and activities planned. (faculty and student exchanges, workshops, meetings) (limited to 250 words)

Bruffaerts' next visit to MIT is tentatively planned for spring 2020 (5 weeks), pending funding. Fedorenko and/or Hosseini will visit Leuven in the summer of 2020 (1-2 weeks). During these visits, the team will focus on data analysis, writing up the results, and preparing new grant applications. During the MIT visit, a workshop focusing on MEG investigations of semantic composition is planned. During the KU Leuven visit, Bruffaerts will organize a symposium about MEG & language. Throughout the duration of the project, Bruffaerts will actively supervise MEG analyses conducted in the Fedorenko Lab and participate in lab meetings. The team will hold skype meetings every 2-3 weeks to touch base on progress.

In this current or proposed collaboration, what other additional activities are planned outside of the seed funding?

Describe additional activities and how they will be funded (NSF, etc.) (limited to 250 words)

Bruffaerts has been and will continue to be involved in training Fedorenko's staff in analyzing MEG data. Bruffaerts and Hosseini will apply to conferences (e.g., Society for Neuroscience, Society for the Neurobiology of Language, Cognitive Neuroscience Society) to present preliminary findings.

Project sustainability

After MISTI seed funding, how will the project continue? (limited to 250 words)

This MISTI-KU Leuven project is scientifically novel and exciting and will enable us to advance our collaboration and allow us to seek extramural funding in the next two years.

A joint publication is currently in preparation. Fedorenko and Bruffaerts plan to submit two grants (one to NSF/NIH, and one to a Belgian or EU funding agency) for MEG projects that would build on the foundation laid by the studies described above. Bruffaerts is currently the only Belgian researcher using MEG in language research and aims to expand this MEG research line at KU Leuven in the near future. In the US, only a handful of labs (e.g., at NYU, CMU, MGH) use MEG to study language processing. Fedorenko's involvement in this project will add MIT to this list.

Are you applying for any other seed or research funds within MIT for this project?

No

Are you applying for any other seed or research funds outside MIT for this project?

Yes

Which one(s)? List the name of the grant/fund/program that you are applying for/applied for and the name of the agency/fund providing the funds.

RB is supported by a postdoctoral fellowship of the Research Foundation Flanders which covers her salary. The current fellowship extends until Oct 2020, renewal until 2023 will be requested in Dec 2019. Acquisition of MEG data in Dutch-speaking volunteers was covered by a Research Grant from the Research Foundation Flanders awarded to RB.

Page 3: Proposal Budget

Budget

Note: departmental allocations may apply; please refer to your MIT department.

MIT applicant(s)

Faculty or research scientist travel

(For travel abroad)

Scroll right to complete all required fields

	Team member	Status	Approximate travel dates and length	Activities planned and location	Funds requested in USD (do not use commas or decimal points)	Add row?
1	Ev Fedorenko	Faculty	2 weeks	writing papers/grants, invited symposium	5900	
Total amount requested:						5900.0

Estimated total number of trips planned for all MIT faculty/researchers (going to partner location):

1

MIT student(s)

MIT student travel

(For travel abroad)

Scroll right to complete all required fields

	Student	Year (U or G)	E-mail	Approximate travel dates and length	Activities planned and location	Funds requested in USD (do not use commas or decimal points)	Add row?
1	Eghbal Hosseini	grad	ehoseini@MIT.EDU	2 weeks	analysis, writing paper	5900	
Total amount requested:							5900.0

Estimated total number of trips planned for all MIT students (going to partner location):

1

Estimated total number of all MIT students (undergraduate and graduate) who will participate in this project:

2

International collaborator(s)

International faculty and research scientists travel

(For travel to MIT)

Scroll right to complete all required fields

	Collaborator	Approximate travel dates and length	Activities planned	Funds requested (do not use commas or decimal points)	Add row?
1	Rose Bruffaerts	May 2020 (5 weeks)	analysis, writing, teaching	13472	✓
2					✗
Total amount requested:	13472.0				

Estimated total number of trips planned for all international Collaborators (going to MIT):

1

International student(s)

International student(s) travel

(For travel to MIT)

Scroll right to complete all required fields

	Collaborator	Approximate travel dates and length	Activities planned	Funds requested (do not use commas or decimal points)	Add row?
1					
Total amount requested:	0				

Estimated total number of trips planned for all international students (going to MIT):

0

Estimated total number of all International/non-MIT students (undergraduate and graduate) who will participate in this project:

0

Non-travel related items (meeting and/or workshop)

Not travel related items

Scroll right to complete all required fields

	Activities planned (and location)	Approximate dates and length	Funds requested (do not use commas or decimal points)	Add row?
1	Symposium MEG investigations of semantic composition (MIT)	May 2020, 1 day	1000	✓
2	Symposium: MEG in language research (Leuven)	Summer 2020, 1 day	1000	✗
Total amount requested			2000.0	

Total funds requested in USD:

27272.00

Overhead amount (10%)

2727.20

Total USD with Overhead:

29999.20

**When you've completed the application, click the green "Mark as Complete" button below.
When you are ready to submit your proposal, click "Review & Submit" in the left-hand taskbar.**